S&P GLOBAL X CRISIL CAMPUS HACKATHON 2026 · CASE STUDY

TREMOR

A headline names one company.
We show a bank the loss on its whole book, including the
obligors the headline never named, with a range and a trail to the sources.

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Engine: sentiment · event class · impact | Module B: event-driven portfolio stress test | Module A: rebalancer (second subscriber)

Problem and approach

The gap

Text signals are generic (one score per ticker). Reposts count as evidence. Rumours fire alerts. Stress tests are periodic and detached from news. No uncertainty, no audit trail.

The question

"If this headline is real, what does it do to OUR book, how sure are we, and can I show the auditor exactly how we got the number?"

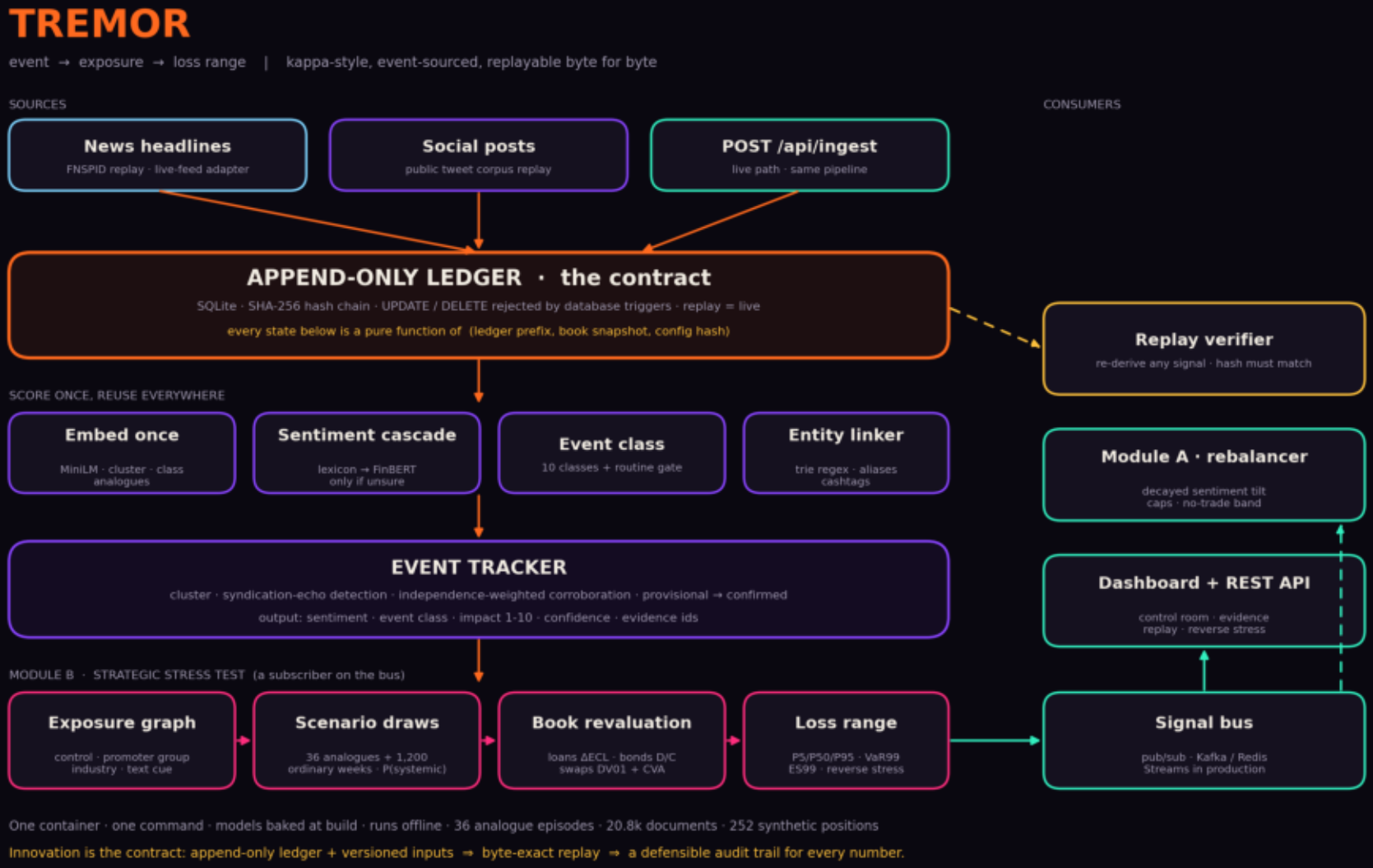
One claim

Every risk signal is a loss range for a specific book, derived from historical analogues, traceable to its sources, and exactly reproducible.

Approach

text → event (clustered, syndication-aware, independence-weighted) → exposure graph (control · promoter group · industry · text cue) → scenario draws from 36 documented historical episodes + ordinary weeks (P(systemic)) → Monte-Carlo revaluation of a synthetic wholesale book (loans, bonds, swaps) → P5/P50/P95 loss, reverse stress, evidence lineage, byte-exact replay.

System design



Implementation highlights

Append-only ledger

SQLite + SHA-256 hash chain; UPDATE/DELETE rejected by triggers. Replay = live path.
Why: audit trail and exact reproducibility.

Score once

One MiniLM embedding feeds clustering, class, novelty and analogues. Sentiment cascade: lexicon first, FinBERT only when unsure.

Events, not articles

Echo detection (syndicated copies are not independent), publisher-independence weights, social volume capped. provisional → confirmed.

Exposure graph

Parent / subsidiary, promoter group, industry peers and rule-based text cues. Reaches the obligors the headline never named.

Module B stress

Credit triangle (spread ≈ PD×LGD); loans ΔECL, bonds duration/convexity, swaps DV01 + CVA. Vectorised Monte Carlo; reverse stress.

Engineering

FastAPI · Pydantic contracts · pytest (36 tests, incl. replay exactness) · ruff · CI · one-container Docker, models baked, offline at runtime.

Key results (measured, out of sample)

85%

of realised market moves inside the predicted 90% band
150 intervals · chronological · no look-ahead

+7.5%

pinball skill vs drawing every event from the crisis library
+18.3% vs ordinary weeks

76%

FinBERT calls avoided by the cascade
accuracy 74.8% vs FinBERT 65.5%, lexicon 73.7%

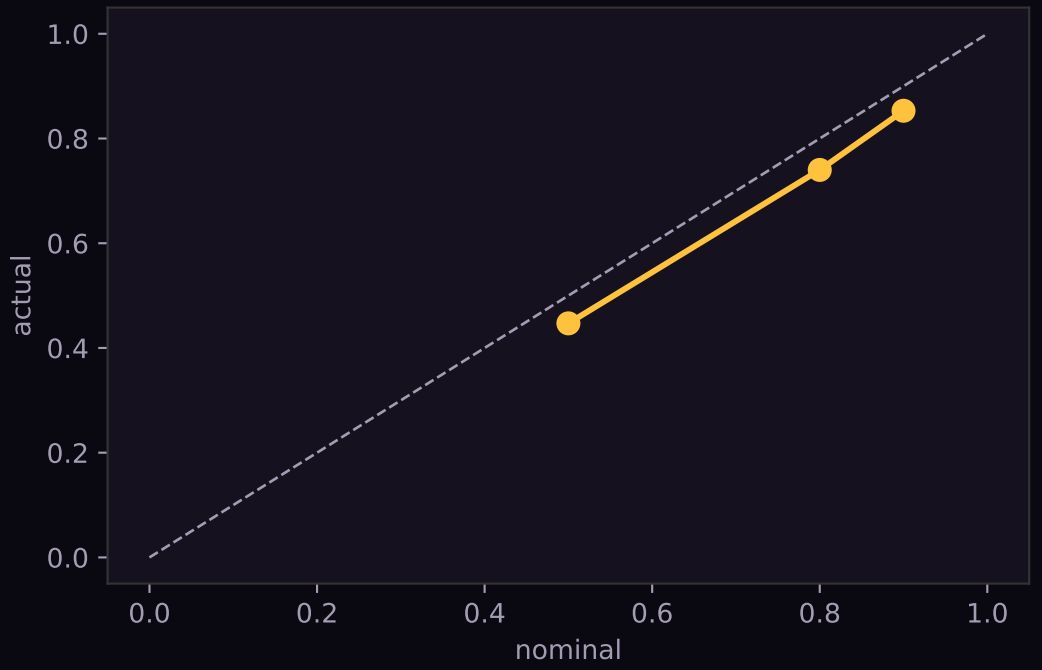
20,836 → 6,123

docs → events: routine/spam filtered, echoes collapsed
32% of news were syndicated copies · 5.1 ms/doc

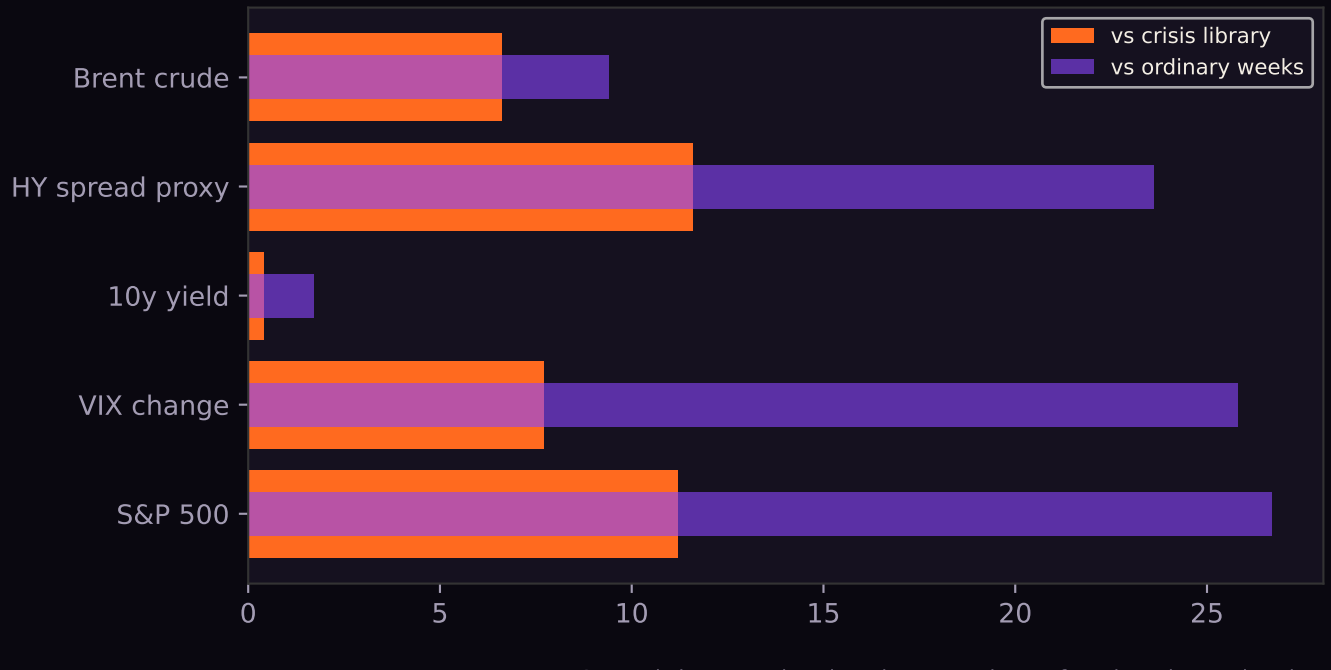
ρ 0.23

Impact vs realised |abnormal return|
n=103 single-name events, p=0.022

Calibration: nominal vs actual coverage



Skill by target (%, higher is better)



6 · WHY IT MATTERS

Domain impact

Time to exposure

From headline to a book-specific loss range in seconds, with the obligors and paths that carry it. Triage first, analyst judgement second.

Fewer false alarms

Syndicated copies and tweet volume never count as confirmation. Provisional signals carry wide ranges that tighten as independent sources agree.

Audit-ready

Every number traces to source documents and re-derives byte for byte from the ledger, the book snapshot and the config hash.

Positioning for risk teams

Complements, does not replace, surveillance and supervisory stress testing: it is the fast, transparent first estimate that tells an analyst where to look. Regulatory scenarios are periodic and fixed; static group-exposure aggregation exists, but it does not react to a headline, weigh evidence or give a range. Module A (rebalancer) was added as a second bus subscriber without changing the engine, showing how consumers plug in.

Limitations and next steps

What did not work

Sharper analogue retrieval (text / class / severity) did NOT beat the unconditional history at n=36, so we keep it soft and use what did work: P(systemic | class).
Impact is a transparent triage score, not a calibrated forecast.
Promoter-group contagion cannot be backtested (1 episode).

Assumptions

News is day-resolution (no minute-level lead/lag claims).
Synthetic book and ratings; real names are labels only.
Credit spreads are ETF proxies; credit triangle, LGD 45%.
Horizon: 5-day peak excursion. Ranges are illustrative.

Next steps

Grow the episode library (hundreds) so conditioning can be tested properly · intraday timestamps and a licensed live feed · multilingual local-press ingestion · GLEIF parent-child data for real group graphs · Kafka / Redis Streams behind the same bus interface.